

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1

Year: 2011

Subject: 394171 Mathematics I

Section: 15-16

Date: 25 July 2011

Time: 08.00-10.00

Name _____ ID _____ Field of study _____

Instructions

1. Cheating will result in failure of all classes registered for the current semester. Student who are caught cheating will also be denied registering for the following semester.
 2. No documents are allowed to be taken out of the examination room.
 3. Textbooks are not allowed.
 4. Calculation devices and dictionary are allowed.
 5. The examination has 5 pages (including this page), 7 questions and a total score of 70 points.
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Question 1 Given $U = \{1, 2, 3, 4, 5, 6, 7, 8, \dots\}$

$$A = \{3, 6, 9, 12, 15, 18, 21, 24, 27, \dots\} \text{ and}$$

$$B = \{\{3, 6\}, \{9, 12, 15\}, 18, 21, 24, 27, \dots\}$$

How many elements for the set $(A - B) \cup (B \cap A')$? (10 points)

Question 2 Given $A = \{a\}$, $B = \{a, b\}$, $C = \{b, c, d\}$ and $D = \{a, b, c, d\}$.

Find the following sets (10 points)

a. $(D - A) \cup (D - C)$

2. $\{D - A\} \cap C$

Question 3 Let $A = \{ \emptyset, 0, 1, \{0,1\} \}$ and $B = \{ \emptyset, \{\emptyset\}, \{0, \{0,1\}\}, \{0, \{1\}\} \}$.

$P(A)$ is a power of the set A. How many elements for the set $P(A) - B$? (10 points)

Question 4 Given $n(A) = 50$, $n(B) = 40$, $n(C) = 30$, $n(A \cap B) = 15$, $n(B \cap C) = 13$, $n(A \cap C) = 17$ and $n(A \cup B \cup C) = 80$.

Find the numbers of elements for the set $(A \cap B) - C$? (10 points)

Question 5 if a, b, c , and d are real numbers with $0 < a < b$ and $0 < c < d$.

Prove that $\frac{a}{d} < \frac{b}{c}$. (10 points)

Question 6 Given $A = \{x \in \mathbb{R} \mid 3x^2 + 2 < -5x\}$ and

$$B = \{x \in \mathbb{R} \mid \frac{1+2x}{3-x} < 0\}$$

Find $A' \cap B$ (10 points)

Question 7 Given R is a real number. Find the set of all solutions of the inequality

$$\frac{x^4}{x^2-1} \geq x^2+1 \quad (10 \text{ points})$$

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